# Pythagorean Theorem

*Teacher's Edition*

## The Period

**MA.8.GR.1.1** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

**Must** Find a missing leg or the hypotenuse of a right triangle from $a^2 + b^2 = c^2$, the hypotenuse found first; an irrational answer exact and, when asked, to the nearest tenth.

**Must not** Stories (6.04); the coordinate plane (6.05); radicands over 225 or with a square factor; 3-D.

**Target** I can find a missing side of a right triangle — the hypotenuse, or a leg.

**MTR** **MTR.6.1** — Notes III and boards 3–4: every leg is checked against c · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.5.1** — Notes I and board 5: one theorem however the triangle is turned

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 10 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Find the square corner first: the side across from it is c, the longest side.
2. The hypotenuse: square, add, then the root.
3. A leg: subtract the squares — and check it is shorter than the hypotenuse.

## Slide by Slide

1. **Pythagorean Theorem.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes III's move before it has a name: $10^2 - 6^2 = 64 = 8^2$, the missing leg of the 6-8-10 triangle Notes II builds. *1 retrieves 6.01–02 — the Triangle Inequality, the last class (yesterday). 2 retrieves 5.10–11 — a unit rate used once (last week). 3 retrieves 4.05 — a perfect square's root (Unit 4) (two units back). 4 retrieves MA.7.NSO.1.1 — whole-number powers; a missing leg subtracts two squares like this (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. No — 4 + 7 = 11 < 12 2. 81 pages 3. 14 4. 64
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the theorem and the picture.** Find the square corner. The side across from it is c.
6. **Notes.** Square, add, then the root. The root is the last step.
7. **Notes.** A leg is shorter than the hypotenuse. Always check.
8. **Example 1 — Find the hypotenuse.** Ask where the square corner is and which side is c before anyone squares.
9. **Example 1 — Worked.** 49 + 81 = 130, not a perfect square: c = √130, between 11 and 12 (121 < 130 < 144), and 11.40 to two places, so 11.4.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 25 + 121 = 146 and √146 ≈ 12.08, so 12.1 — a board with 16 added the legs; a board with 146 stopped before the root.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Find a leg.** Ask which number is c before anyone squares: 15, the longest — so this time a square is subtracted.
13. **Example 2 — Worked.** 15 is c, so $15^2$ stands alone: $b^2 + 64 = 225$, $b^2 = 161$, b = √161 ≈ 12.69, so 12.7 — and 12.7 is shorter than 15, as a leg must be.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 121 − 36 = 85 and √85 ≈ 9.22, so 9.2 — a board with 12.5 added the squares (√157), longer than the hypotenuse.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** A right triangle has legs of 9 cm and 12 cm. How long is the hypotenuse? → ***15 cm*** *21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]*

**18–19. Board 2.** A right triangle has legs of 4 inches and 9 inches. How long is the hypotenuse, to the nearest tenth? → ***√97 in ≈ 9.8 in*** *13 — added the legs [MA.8.GR.1.1]; 9.9 — rounded twice [Notes II: round once, at the end]*

**20–21. Board 3.** A right triangle has a hypotenuse of 25 ft and one leg of 20 ft. How long is the other leg? → ***15 ft*** *32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]*

**22–23. Board 4.** A right triangle has a hypotenuse of 41 m and one leg of 40 m. How long is the other leg? → ***A — 9 m*** *(B) stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]; (C) subtracted the sides, 41 − 40 [MA.8.GR.1.1: the theorem is about squares]; (D) added the squares instead of subtracting: about 57.3, longer than the hypotenuse [Notes III: for a leg, subtract] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** The right triangle is drawn turned. Its legs are 5 cm and 7 cm. How long is the hypotenuse, to the nearest tenth? → ***√74 cm ≈ 8.6 cm*** *4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]*

**26–27. Board 6.** A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x. Which equation finds x? → ***D — and x = 12 in*** *(A) the sides added, not their squares [MA.8.GR.1.1]; (B) the hypotenuse put with the legs [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (C) doubled the 20 instead of squaring it — $20^2$ means 20 × 20, not 2 × 20 [Notes I: the theorem is about squares] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** A right triangle has a hypotenuse of 14 m and one leg of 9 m. How long is the other leg, to the nearest tenth? → ***√115 m ≈ 10.7 m*** *16.6 — added the squares, longer than the hypotenuse [Notes III]*

**30–31. Board 8.** A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars. How long is its longest side, to the nearest tenth? → ***√185 in ≈ 13.6 in*** *The 6 dollars is not needed. 19 — added the legs [MA.8.GR.1.1]; 'cannot tell' — the longest side is the hypotenuse [Notes I]*

**32–33. Board 9.** Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm. She writes $\sqrt{(5^2 + 13^2)} \approx 13.9$. What went wrong? How long is the leg? → ***She added the squares; a leg must be shorter than 13 cm. Subtract: the leg is 12 cm.*** *'13.9 is fine' — no check against the hypotenuse [Notes III]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: the hypotenuse, whole then to the tenth; 3–4 a leg, whole then the vote. The pattern breaks at 5 — the triangle turned — and 6 asks only for the equation; 7 is a leg to the tenth; 8 has the extra number; 9 is the check in writing. The independent set: a whole hypotenuse, one to the tenth, a whole leg from a big hypotenuse, a leg to the tenth, a turned triangle, the equation and its value.

## Misconceptions to Watch

**Board 1.** 21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]

**Board 2.** 13 — added the legs [MA.8.GR.1.1]

**Board 3.** 32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]

**Board 4.** stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]

**Board 5.** 4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]

**Board 6.** the sides added, not their squares [MA.8.GR.1.1]

**Board 7.** 16.6 — added the squares, longer than the hypotenuse [Notes III]

**Board 8.** 19 — added the legs [MA.8.GR.1.1]

**Board 9.** '13.9 is fine' — no check against the hypotenuse [Notes III]

**Materials** Whiteboards and markers. $a^2 + b^2 = c^2$ on the board for the period (memorized by the test). Calculators allowed, printed nowhere.